

Effects of Diode Low-Level Laser Therapy of 810 Nm on Pulpal Anesthesia of Maxillary Premolars: A Double-Blind Randomized Clinical Trial

 Iman SOKOUTI EMAMZADEH HASHEMI,  Dina MALEKI,  Seyyed EBRAHIM SEYYED MONIR,  Arash EBRAHIMI,  Rasool TABARI,  Elnaz MOUSAVI

Please cite this article as: Sokouti Emamzadeh Hashemi I, Maleki D, Ebrahim Seyyed Monir S, Ebrahimi A, Tabari R, Mousavi E. Effects of Diode Low-Level Laser Therapy of 810 Nm on Pulpal Anesthesia of Maxillary Premolars: A Double-Blind Randomized Clinical Trial. Eur Endod J 2021; 155-9

From the Department of Endodontics (I.S.E.H., E.M. ✉ elnaz32@gmail.com), Faculty of Dentistry, Guilan University of Medical Sciences, Rasht, Iran; Dental Student (D.M.), Student Research Committee, Faculty of Dentistry, Guilan University of Medical Sciences, Rasht, Iran; Department of Periodontics (S.E.S.M.), Faculty of Dentistry, Guilan University of Medical Sciences, Rasht, Iran; Master of Science in Lasers in Dentistry (A.E.), Rheinisch-Westfälische Technische Hochschule Aachen University, Aachen, Germany; Social Determinants of Health Research Center (R.T.), Guilan University of Medical Sciences, Rasht, Iran

Received 15 April 2020,
Accepted 02 August 2020

Published online: 25 May 2021
DOI 10.14744/eej.2020.41636

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HIGHLIGHTS

- Fear and anxiety of dentistry prevent patients from visiting dental clinics.
- Laser have pain-relieving and anti-inflammatory effects.
- If the purpose is merely to raise the pain threshold, the use of laser is effective.
- Laser lessens the patients' fear of dental procedures.

INTRODUCTION

Fear and anxiety of dentistry procedures are influential factors that prevent patients from visiting dental clinics at the right time, which is largely resulted from the experience of pain during dental procedures; there is a significant association between painful experiences during dental treatments in childhood and anxiety and fear of dental procedures in adulthood (1, 2).

The dental pulp is an oral tissue composed of multiple neural elements. In modern dentistry, local anesthesia of pulpal tissue during treatment has been proven to block the pathway of pain sensation (or transduction) (3). But, drug injection itself can intensify anxiety and fear, particularly in younger patients (4). However, according to some studies, the effectiveness of anesthesia has not been completely effective, which together with the fear and anxiety of drug injection, over-shadows the continued application of this procedure (5). Besides these, other complications and/or consequences of this procedure include needle breakage, paresthesia, facial nerve paralysis, hematoma, systemic consequences on the cardiovascular system, liver disease, allergies, drug overdose, and limitations of local anesthesia in pregnant women and some of the patients with systemic complications (6-9).

Laser therapy is currently at the experimental and research stages. Remarkable results have been achieved using this procedure in the treatment of conditions such as dentine hypersensitivity, temporomandibular joint (TMJ) syndrome, inferior alveolar nerve (IAN) paresthesia caused by surgical removal of 3rd molars, trigeminal neuralgia, herpes labialis, aphthous ulcers and inflammation after chemotherapy (10-15). The advantages of this procedure include diminished bleeding during and after treatment (and thereby efficient access and monitoring), pain-relieving and anti-inflammatory effects, stress-less intervention, diminished gingival swelling through closing lymphatic vessels, no need for sutures and no scarring after surgery (16-18).

According to some studies, the process of pulpal anesthetic may be affected by diode laser irradiation (19). Some researchers have also stated that Photobiomodulation Therapy (PBMT) can affect both C and A δ nerve fibers, whilst other studies have reported that only C fibers are affected (20). Also, a literature review has presented reliable evidence regarding the mechanism of nerve function blockage using PBMT by diode laser for pain and anesthesia (21).

Therefore, and given the easy access to the 810 nm diode laser, this study was designed and conducted to investigate effects of diode low-level laser therapy (LLLT) of 810 nm on pulpal anesthesia of maxillary premolars regarding the importance and efficient implementation of anesthesia in dental procedures and the development of lasers, which is evident in most dimensions of dental treatments, as well as the contradictory results of previous studies on the performance and effectiveness of LLLT at 810 nm on dental anesthesia.

This study was designed with the null hypothesis that the induction of pulpal anesthesia to the maxillary premolars would be similarly effective in both the control and the treatment groups.

MATERIALS AND METHODS

This is a parallel- group, prospective, double-blind randomized trial. Ethical clearance was obtained from University Ethics Committee (UEC). CONSORT 2010 guidelines were applied. Eligibility criteria and method were not changed during the study. All patients were also asked to sign the informed consent form.

The sample size was calculated to be 20 according to Bahramian (2011) with a standard ratio of 1:1 ($P=0.85$), but in total 25 patients were included regarding the probable dropouts in the study population.

Participants were patients with at least two permanent maxillary premolars without dental caries and repair and/or other dental defects, and also without cardiac arrhythmia or a history of using a cardiac pacemaker, malignant tumors, epilepsy and pregnancy, local infection, blood pressure diseases, photophobia, dental orthodontics, and dental hypersensitivity. Patients who did not attend the second session were excluded from the study.

Premolars were initially isolated with a cotton roll. The treatment randomly performed on the first chosen premolars as described

earlier. The electric pulp testing (EPT) device (Epell's Parkell, Inc, Edgewood, New York, USA) was utilized to assess potential alterations in action potentials at the nerve terminals before and after treatment. The EPT electrode was placed on the buccal cusp of the premolar teeth and prophylaxis paste was used at the point of contact of the device's tip to facilitate the flow of electrical energy (20). Each patient was asked to immediately remove his/her hand over the metal handle of the EPT device after feeling tingling, burning, itching, moderate or slight electric shock in teeth, and pain. And the EPT number was recorded by the operator. To ensure the accuracy of the results, the patient's sense of EPT was examined on a different tooth previous to the study under precise isolation to introduce the patient with the sensations experienced during the study.

Two symmetrical teeth on the left and right sides were tested to evaluate sham laser and low power diode laser at 810 nm. All patients underwent both laser and sham laser treatments randomly. A coin dropping method was adopted to determine which of the laser and sham laser should be randomly applied firstly for each patient. Then a coin dropped for the second time to determine which tooth would undergo the first treatment using laser or sham laser.

To ensure blindness, the laser light-emitting medium was removed from the device in the placebo mood, but both the operator and the patient were hearing laser emission warning sound, and hence the patient could not recognize the laser emission. Except for the operator, neither the patient nor the recorder (who was someone other than the operator) had any information about the type of laser probe utilized.

The performance of a specialized low-power laser (Konftec, Corporation, New Taipei City, Taiwan) was confirmed once on buccal surface and once on the palatal surface of the tooth perpendicular to the longitudinal axis and the intermediate line of the hypothetical line between the tip of the cusp and the cervical line of the given tooth at 810 nm, and under constant power of 200 MW, the irradiation time of 30 s, and a cross-sectional area 0.5 cm². The total energy transferred or the dose of laser irradiation to the tooth was equivalent to 6 J. Also, the energy density for each irradiation and in total was respectively 12 J/cm² and 24 J/cm². The laser was applied following safety principles of international standards, such as the installation of a patient warning sign at the entrance to the room during irradiation and the use of radiation shielding glasses particularly at 810 nm.

Pulpal reactions to EPT were assessed immediately under foregoing considerations following laser and sham laser treatment (15, 16). Laser and sham laser treatments were carried out in two different sessions with a one-week interval to ensure avoiding interfere with the biochemical effects of the laser in reducing or increasing the potential secretion of various mediators and enzymes such as prostaglandin, bradykinin, etc. and consequently, the potential false placebo results.

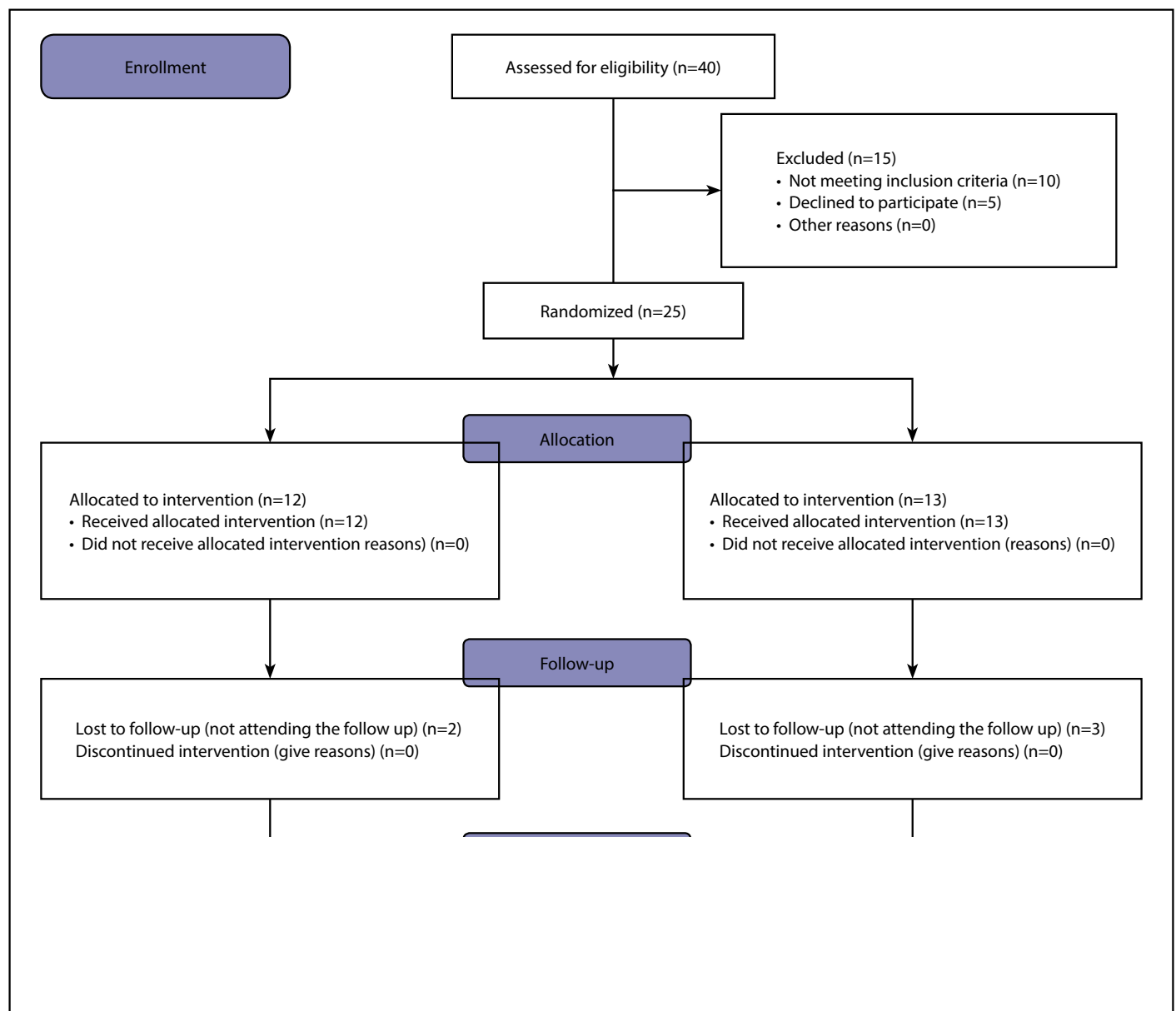


Figure 1. Consort 2010 flow diagram

density of dental pulp.

CONCLUSION

The results of this study confirmed that laser irradiation at 810 nm can significantly reduce pain during EPT assay in patients;

however, these results should be taken with caution since the sample size was small which is a limitation of the current study. There was no correlation between the capacity of the laser at 810 nm to induce anesthesia and gender and age and this wavelength can be used to induce anesthesia. Since pain control is a challenging task in dentistry, laser beams with indicators in the inhibitory phase markers may be an alternative factor in future dental treatments, according to the results of this study. Also, in cases where there is no need for deep anesthesia and the purpose is merely to raise the pain threshold, the use of laser considering indices of this study may eliminate or reduce the need for using topical anesthetics in dentistry, but more research is needed to verify this theory.

Disclosures

Conflict of Interest: Authors declare no conflict of interest.

Ethics Committee Approval: This study was approved by the Guilan University of Medical Science. [IR.GUMS.REC.1397.439]

Peer-review: Externally peer-reviewed.

Financial Disclosure: Authors declare no financial support.

Authorship contributions: Concept – E.M., S.E.S.M.; Design – E.M., I.S.E.H., A.E.; Supervision – E.M., I.S.E.H.; Funding – E.M., S.E.S.M.; Materials – E.M., S.E.S.M.; Data collection &/or processing – E.M., I.S.E.H., A.E., D.M.; Analysis and/or interpretation – E.M., A.E., D.M.; Literature search – E.M.; Writing – E.M., D.M.; Critical Review – E.M., D.M., S.E.S.M.

REFERENCES

- Ngan P, Kess B, Wilson S. Perception of discomfort by patients undergoing orthodontic treatment. *Am J Orthod Dentofacial Orthop* 1989; 96(1):47–53. [CrossRef]
- Oliver RG, Knapman YM. Attitudes to orthodontic treatment. *Br J Orthod* 1985; 12(4):179–88. [CrossRef]
- Chu WS, Park SH, Ahn DK, Kim SK. Effect of local anesthesia on pulpal blood flow in mechanically stimulated teeth. *J Kor Acad Cons Dent* 2006; 31(4):257–62. [CrossRef]
- Aggarwal K, Lamba AK, Faraz F, Tandon S, Makker K. Comparison of anxiety and pain perceived with conventional and computerized local anesthesia delivery systems for different stages of anesthesia delivery in maxillary and mandibular nerve blocks. *J Dent Anesth Pain Med* 2018; 18(6):367–73. [CrossRef]
- Badr N, Aps J. Efficacy of dental local anesthetics: A review. *J Dent Anesth Pain Med* 2018; 18(6):319–32. [CrossRef]
- Amarillas-Escobar ED, Toranzo-Fernández JM, Martínez-Rider R, Noyola-Frías MA, Hidalgo-Hurtado JA, Serna VM, et al. Use of therapeutic laser after surgical removal of impacted lower third molars. *J Oral Maxillofac Surg* 2010; 68(2):319–24. [CrossRef]
- Shirani AM, Gutknecht N, Taghizadeh M, Mir M. Low-level laser therapy and myofascial pain dysfunction syndrome: a randomized controlled clinical trial. *Lasers Med Sci* 2009; 24(5):715–20. [CrossRef]
- Brugnera A, dos Santos AEC, Bologna ED, Ladalarlo TC. *Atlas of Laser Therapy Applied to Clinical Dentistry*. São Paulo: Quintessence Editoria; 2006.
- Goulart CS, Nouer PR, Mouramartins L, Garbin IU, de Fátima Zanirato Lizarelli R. Photoradiation and orthodontic movement: experimental study with canines. *Photomed Laser Surg* 2006; 24(2):192–6. [CrossRef]
- Sigaroudi AK, Maleki D, Zare H, Maleki D. Is Low Level Laser Therapy Effective for Complications of Mandibular Third Molar Surgery? A Literature Review. *IJSRDMS* 2020 Jul 20 [Epub ahead of print], doi:10.30485/IJSRDMS.2020.229110.1053.
- Capodiferro S. Comment on "efficacy of low-level laser therapy in management of recurrent herpes labialis: a systematic review". *Lasers Med Sci* 2019; 34(4):841. [CrossRef]
- Björdal JM, Lopes-Martins RA, Joensen J, Couppe C, Ljunggren AE, Stergioulas A, et al. A systematic review with procedural assessments and meta-analysis of low level laser therapy in lateral elbow tendinopathy (tennis elbow). *BMC Musculoskelet Disord* 2008; 9:75. [CrossRef]
- Kim SJ, Moon SU, Kang SG, Park YG. Effects of low-level laser therapy after Corticision on tooth movement and paradental remodeling. *Lasers Surg Med* 2009; 41(7):524–33. [CrossRef]
- Yan W, Chow R, Armati PJ. Inhibitory effects of visible 650-nm and infrared 808-nm laser irradiation on somatosensory and compound muscle action potentials in rat sciatic nerve: implications for laser-induced analgesia. *J Peripher Nerv Syst* 2011; 16(2):130–5. [CrossRef]
- Tanboga I, Eren F, Altinok B, Peker S, Ertugral F. The effect of low level laser therapy on pain during dental tooth-cavity preparation in children. *Eur Arch Paediatr Dent* 2011; 12(2):93–5. [CrossRef]
- Orchardson R, Whitters CJ. Effect of HeNe and pulsed Nd:YAG laser irradiation on intradental nerve responses to mechanical stimulation of dentine. *Lasers Surg Med* 2000; 26(3):241–9. [CrossRef]
- Marković AB, Todorović L. Postoperative analgesia after lower third molar surgery: contribution of the use of long-acting local anesthetics, low-power laser, and diclofenac. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2006; 102(5):e4–8. [CrossRef]
- Hagiwara S, Iwasaka H, Hasegawa A, Noguchi T. Pre-Irradiation of blood by gallium aluminum arsenide (830 nm) low-level laser enhances peripheral endogenous opioid analgesia in rats. *Anesth Analg* 2008; 107(3):1058–63. [CrossRef]
- Shanthala BM, Wilson B, Joppan S, Srihari. Current Uses of Diode Lasers in Dentistry. *Otolaryngol* 2017; 7(2):295.
- Ricketts D, Bartlett DW. *Advanced Operative Dentistry E-Book: A Practical Approach*. Elsevier Health Sciences; 2011.
- Khalighi Sigaroudi A, Maleki D, Zare H, Maleki D. Is low-level laser therapy effective for complications of mandibular third molar surgery? A literature review. *IJSRDMS* 2020; 2(3):72–80.
- Corona SA, Nascimento TN, Catirse AB, Lizarelli RF, Dinelli W, Palma-Dibb RG. Clinical evaluation of low-level laser therapy and fluoride varnish for treating cervical dentinal hypersensitivity. *J Oral Rehabil* 2003; 30(12):1183–9. [CrossRef]
- Chan A, Armati P, Moorthy AP. Pulsed Nd: YAG laser induces pulpal analgesia: a randomized clinical trial. *J Dent Res* 2012; 91(7 Suppl):79S–84S.
- Weisleder R, Yamauchi S, Caplan DJ, Trope M, Teixeira FB. The validity of pulp testing: a clinical study. *J Am Dent Assoc* 2009; 140(8):1013–17.
- Dilsiz A, Aydin T, Emrem G. Effects of the combined desensitizing dentifrice and diode laser therapy in the treatment of desensitization of teeth with gingival recession. *Photomed Laser Surg* 2010; 28 Suppl 2:S69–74.
- Machado AC, Viana ÍEL, Farias-Neto AM, Braga MM, de Paula Eduardo C, de Freitas PM, et al. Is photobiomodulation (PBM) effective for the treatment of dentin hypersensitivity? A systematic review. *Lasers Med Sci* 2018; 33(4):745–53. [CrossRef]
- Ueda Y, Shimizu N. Effects of pulse frequency of low-level laser therapy (LLLT) on bone nodule formation in rat calvarial cells. *J Clin Laser Med Surg* 2003; 21(5):271–7. [CrossRef]
- Eslamian L, Borzabadi-Farahani A, Hassanzadeh-Azhiri A, Badiie MR, Fekrazad R. The effect of 810-nm low-level laser therapy on pain caused by orthodontic elastomeric separators. *Lasers Med Sci* 2014; 29(2):559–64. [CrossRef]
- Chow RT, Armati PJ. Photobiomodulation: Implications for Anesthesia and Pain Relief. *Photomed Laser Surg* 2016; 34(12):599–609. [CrossRef]
- Aras MH, Omezli MM, Güngörmüş M. Does low-level laser therapy have an antianesthetic effect? A review. *Photomed Laser Surg* 2010; 28(6):719–22. [CrossRef]